CLINICAL · EAR

Middle ear and temporal bone tumours

Tumours of the middle ear and temporal bone are rare, and two of them must never be missed. The first is a red, pulsating mass behind the drum, a glomus tumour that bleeds torrentially if biopsied in clinic. The second is the chronic, painful, bleeding ear that is really a squamous cell carcinoma.

AT A GLANCE

- Never biopsy a pulsatile red mass.
- CT and MRI separate glomus from carotid and jugular anomalies.
- Check catecholamines; embolise before surgery.
- A chronic bloody painful ear needs biopsy.
- Facial schwannoma: operate only for weakness worse than House–Brackmann III.

1 CHAPTER ONE Overview

Rare, but two are not to be missed.

1.1 The tumours

TUMOURS OF THE MIDDLE EAR AND TEMPORAL BONE

- BENIGN **Paraganglioma (glomus)** tympanicum, jugulare, vagale
- BENIGN **Others** adenomatous neuroendocrine tumour, facial schwannoma, haemangioma, choristoma, endolymphatic sac tumour
- MALIGNANT **Squamous cell carcinoma** the commonest
- MALIGNANT **Others** adenoid cystic carcinoma, adenocarcinoma, rhabdomyosarcoma (children), metastases
- SKULL BASE **Chordoma, chondrosarcoma** petrous apex and clivus

1.2 Why it matters

Both key tumours present to the ear clinic looking like something common: pulsatile tinnitus or a polyp, and a discharging ear.

2 CHAPTER TWO Paraganglioma

The glomus tumour.

2.1 Origin

Paragangliomas arise from neural-crest *glomus bodies* along the tympanic branch of the glossopharyngeal nerve (Jacobson's nerve) and the auricular branch of the vagus (Arnold's nerve), and on the dome of the jugular bulb. They are benign, slow-growing and highly vascular.

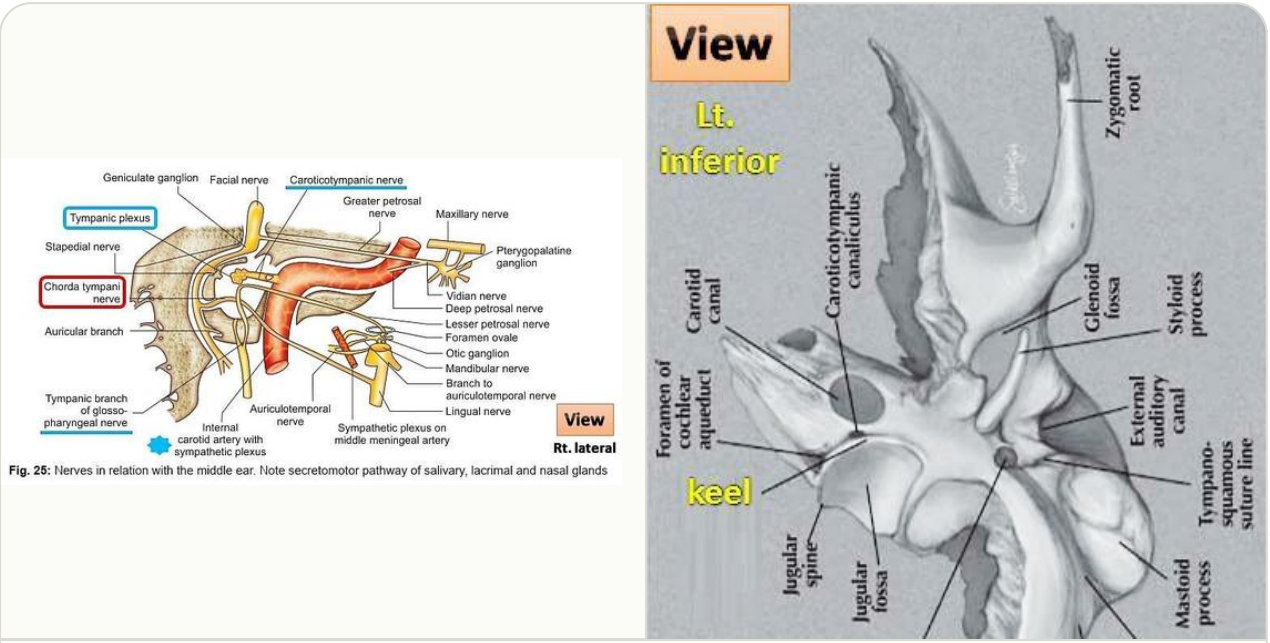


Fig. 25: Nerves in relation with the middle ear. Note secretomotor pathway of salivary, lacrimal and nasal glands

FIGURE 1 The tympanic plexus on the promontory (Jacobson's nerve), and the keel of bone between the carotid canal and jugular fossa.

Anatomy atlas · ENT Steps p. 53, ENT Steps p. 24

2.2 Who gets them

Usually in the fourth and fifth decades, with a strong female predominance (about 5 to 1).

2.3 Pathology

Nests of chief cells (*zellballen*) surrounded by sustentacular cells in a vascular stroma.

2.4 Genetics

Some are familial, from mutations in the succinate dehydrogenase genes. SDHD mutations cause multiple tumours and are expressed only when inherited from the father; SDHB mutations carry a higher risk of malignancy.

WHERE THE BOOKS DIFFER · HOW OFTEN

Dhingra	Lee
About 10% familial, about 3% secrete catecholamines, about 4% metastasise.	Up to 30% familial, 5% or fewer secrete, 2–4% metastasise.

2.5 Types

2.5.1 Glomus tympanicum

Arises on the promontory and fills the middle ear.

2.5.2 Glomus jugulare

Arises in the jugular bulb, eroding upwards into the middle ear and medially to the jugular foramen and skull base.

2.5.3 Glomus vagale

Arises on the vagus below the skull base, presenting in the neck.

2.6 Clinical features

2.6.1 Symptoms

Pulsatile tinnitus (about 80%), conductive hearing loss (about 60%), fullness, and later bleeding from the ear. Jugulare tumours add lower cranial nerve palsies (ninth to twelfth: hoarseness, dysphagia, wasting of the tongue and trapezius) and facial weakness.

2.6.2 Signs

A red mass behind the drum, the *rising sun* when it emerges from the floor of the middle ear. Pneumatic pressure on the drum makes the mass pulsate, then blanch (*Brown's sign*). A bruit may be heard over the mastoid. In advanced cases a bleeding polyp fills the canal.



FIGURE 2 Glomus jugulare: a red mass behind the drum.
Dhingra, fig. 17.1

NEVER BIOPSY IN CLINIC

A red, pulsatile mass behind the drum, or a polyp with pulsatile tinnitus, is a glomus tumour until imaged. Biopsy can cause catastrophic bleeding.

2.6.3 Secreting tumours

Hypertension, headache, palpitations and sweating mean catecholamine secretion: measure urinary or plasma metanephrines before any anaesthetic.

2.7 Investigations

2.7.1 CT

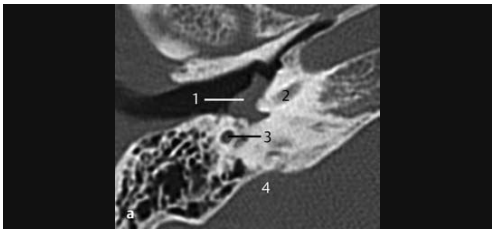
Erosion of the *caroticojugular spine* separates a jugulare tumour from a tympanicum. CT also excludes an aberrant carotid artery and a high jugular bulb.

2.7.2 MRI

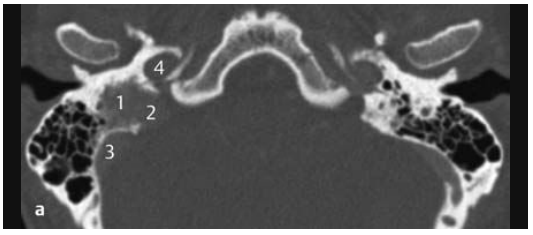
A *salt-and-pepper* appearance from flow voids; defines intracranial and neck extension.

IMAGING IN THE APP 2 cases

Glomus tympanicum and glomus jugulare on CT and MRI.



Glomus Tympanicum (Paraganglioma)
Tumours · 11 images



Glomus Jugulare Tumor (Paraganglioma)
Tumours · 11 images

2.7.3 Angiography

Shows the feeding vessels (mainly the ascending pharyngeal artery), looks for multiple tumours, and allows embolisation. Balloon occlusion testing if the carotid may need to be sacrificed.

2.8 Staging

GLASSCOCK-JACKSON CLASSIFICATION

STAGE	GLOMUS TYMPANICUM	GLOMUS JUGULARE
I	Limited to the promontory	Jugular bulb, middle ear and mastoid
II	Fills the middle ear	Extends under the internal auditory canal
III	Into the mastoid	Into the petrous apex

STAGE	GLOMUS TYMPANICUM	GLOMUS JUGULARE
IV	Through the drum into the canal, or beyond	Beyond the apex into the clivus or infratemporal fossa

The *Fisch* classification grades A (middle ear), B (tympanomastoid), C (infralabyrinthine and apex, C1 to C4 by carotid involvement) and D (intracranial).

2.9 Differential diagnosis

RED, WHITE OR BLUE BEHIND THE DRUM

COLOUR	THINK OF	KEY TEST
Red, pulsatile	Glomus tumour	CT and MRI
Pink, anterior	Aberrant internal carotid artery	CT, MR angiography
Blue, inferior	High jugular bulb	CT, MRI
Blue with dark fluid	Cholesterol granuloma	CT, MRI
White	Congenital cholesteatoma	CT

2.10 Treatment

2.10.1 Surgery

The treatment of choice for most, after embolisation 24 to 48 hours before and alpha-blockade if the tumour secretes. Small tympanicum tumours whose whole margin can be seen are removed through the canal; larger tumours need hypotympanic, extended facial recess or mastoid-neck approaches; jugulare tumours need the Fisch infratemporal fossa approach.

OPERATIVE GUIDES IN THEATRE 3 operations

Glomus Tumors
Glomus tumours

Resection of the Temporal Bone
Temporal bone resection

Tumors of the Facial Nerve
Tumours of the facial nerve

2.10.2 Radiotherapy

Stereotactic radiosurgery or fractionated radiotherapy arrests growth in about 90 to 94%, though the tumour remains. For the elderly, the unfit, residual disease and tumours where surgery would cost lower cranial nerves.

2.10.3 Observation

Slow-growing tumours in elderly patients may simply be watched.

PEARLS

- A glomus tympanicum is only certain when its whole rim is visible; otherwise image.
- Pain is more typical of carcinoma than of a glomus tumour.

3 CHAPTER THREE Squamous cell carcinoma

The chronic ear that bleeds.

3.1 Who gets it

Adults of 40 to 60, about three-quarters with a long history of chronic ear discharge.

3.2 Clinical features

Bloodstained, foul discharge; deep, boring pain, especially at night; friable granulation or a polyp that bleeds on touch; and later facial palsy, vertigo, trismus and a neck or parotid mass. It is easily mistaken for chronic otitis media or otitis externa.

3.3 Investigations

Biopsy of any non-healing granulation in the canal or middle ear. CT of the temporal bone and neck, MRI for soft tissue and dura, and chest imaging.

3.4 Treatment

- **Limited to the canal without bone involvement:** sleeve resection of the canal.
- **Bony canal involved:** lateral temporal bone resection.
- **Middle ear involved:** subtotal temporal bone resection.
- **Parotidectomy and neck dissection** according to spread.
- **Postoperative radiotherapy** for most stages beyond the earliest, with chemotherapy for advanced disease.

Outlook is poor once the middle ear and beyond are involved. The dura usually resists invasion.

CHAPTER FOUR

4 Other tumours

Rarer, each with a clue.

4.1 Rhabdomyosarcoma

The commonest malignant ear tumour in children: a polyp in a child's ear with early facial palsy. Treated with chemoradiation, with resection when feasible.



FIGURE 3 A four-year-old child with rhabdomyosarcoma of the right middle ear and mastoid, with facial palsy on the same side.

Dhingra, fig. 17.2

4.2 Endolymphatic sac tumour

A locally aggressive papillary tumour of the sac, sporadic or with von Hippel-Lindau disease (about 11% of patients, often bilateral). It mimics Ménière's disease, or causes sudden hearing loss from bleeding into the inner ear. Removed with the dura around the sac; VHL patients are screened by MRI every one to three years.

4.3 Facial nerve schwannoma and haemangioma

A schwannoma causes slowly progressive facial palsy or hearing loss; a haemangioma near the geniculate ganglion causes a palsy out of proportion to its small size and should be removed early. Facial schwannomas arise most often near the geniculate ganglion, and many patients keep good facial function for years. Because removal means sacrificing and grafting the nerve, and the best result after a graft is House–Brackmann III, surgery is reserved for weakness worse than grade III; enlarging tumours with good function are treated by radiosurgery (control in

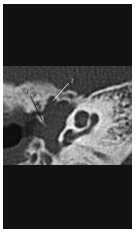
about 90% at five years, with function preserved in 90%) or by bony decompression of the canal.

4.4 Adenomatous neuroendocrine tumour

A benign middle ear tumour presenting as a mass behind an intact drum with conductive loss.

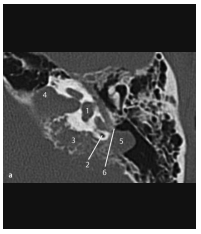
IMAGING IN THE APP 5 cases

Middle ear adenoma, endolymphatic sac tumour, canal carcinoma, fibrous dysplasia and apical pseudotumours.



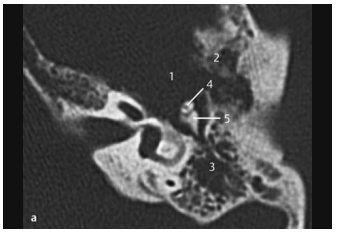
Adenoma of the Middle Ear

Tumours · 1 image



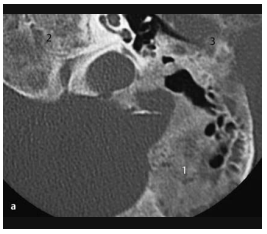
Endolymphatic Sac Tumour

Tumours · 8 images



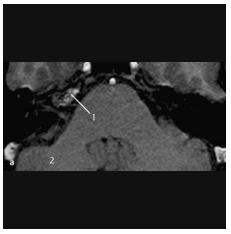
Malignancy of the External Auditory Canal Squamous Cell Carcinoma

Tumours · 2 images



Fibrous Dysplasia of the Temporal Bone and Skull Base

Bone disease · 7 images



Pseudotumors in the Apex of the Petrous Bone

Tumours · 12 images

4.5 Chordoma and chondrosarcoma

Chordoma arises in the midline clivus in older patients, with headache and sixth nerve palsy; it is bright on T2 MRI and contains physaliferous cells. Chondrosarcoma arises off the midline at the petroclival junction. Both are treated by maximal resection followed by high-dose or proton radiotherapy.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 On call

A MASS OR GRANULATION IN THE EAR

1. Pulsatile red mass: do not biopsy; arrange CT and MRI.
2. Non-healing, bleeding granulation in a chronic ear: biopsy it.
3. Hypertension, sweating or palpitations with a glomus tumour: check catecholamines before any anaesthetic.

5.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Origin of glomus tumours:** Jacobson's and Arnold's nerves, jugular bulb.
- **Signs:** rising sun, Brown's sign, pulsatile tinnitus.
- **CT:** caroticojugular spine; aberrant carotid and high bulb.
- **Staging** and treatment options.
- **Squamous carcinoma:** who, symptoms, and the types of temporal bone resection.

5.3 Common pitfalls

COMMON PITFALLS

- Biopsying a red mass behind the drum in clinic.
- Treating a bleeding, painful ear as chronic otitis media for months.
- Forgetting catecholamines before surgery.

Written for this app from Oxford Textbook of Otolaryngology ch. 18, Oxford Specialist Handbook ch. 19, Oxford Handbook of ENT ch. 8, Dhingra ch. 16, 17 and K. J. Lee ch. 24. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · TUMOURS

Glomus Tympanicum (Paraganglioma)

IMAGING FINDINGS

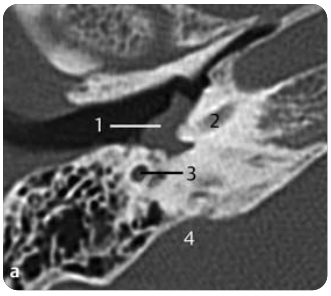
- Small enhancing mass on the cochlear promontory
- Floor of the middle ear intact
- Flow voids only in larger tumours

TOP DIFFERENTIALS

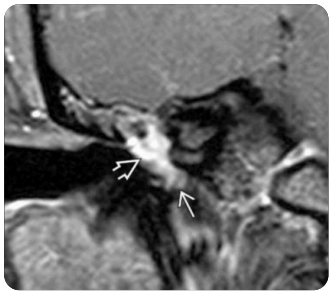
- Glomus jugulare (floor eroded, from the jugular foramen)
- Aberrant internal carotid artery
- Dehiscent jugular bulb

CHECKLIST

- Check the hypotympanic floor and jugular bulb
- Ossicles and extension into the mastoid



CT, axial. On this axial slice at the level of the hypotympanum, a smooth-bordered lesion (1) is seen, without any destruction of the cochlea (2) or…



Coronal T1 C+ FS MR demonstrates a large GTP filling the middle ear cavity. The floor is intact, separating the tumor from the jugular bulb below.…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1064.

TEMPORAL BONE & SKULL BASE · TUMOURS

Glomus Jugulare Tumor (Paraganglioma)

IMAGING FINDINGS

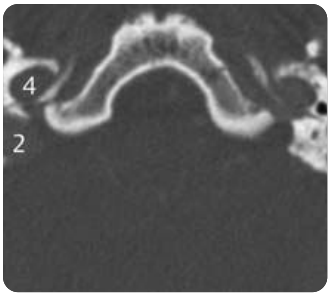
- Jugular foramen mass with permeative, moth-eaten bone erosion
- "Salt and pepper" on T1 with intense enhancement
- Spread superolaterally into the middle ear

TOP DIFFERENTIALS

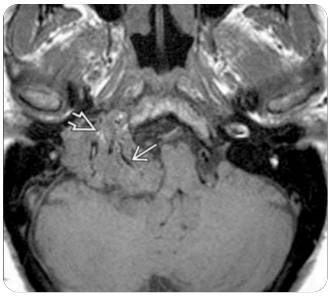
- Jugular foramen schwannoma (smooth scalloping)
- Meningioma (sclerosis, dural tail)
- Dehiscent jugular bulb or metastasis

CHECKLIST

- Carotid, jugular vein and lower cranial nerves
- Multicentric paragangliomas: scan the neck



Patient with pulsating tinnitus, without any pathology behind the eardrum.
Dunnebier, Imaging for Otolaryngologists, Fig. 5.20 a



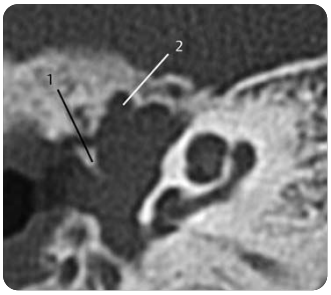
Axial T1WI MR shows a large mass arising from the jugular foramen with multiple areas of salt & pepper. The "salt" represents blood products or…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 892.

MIDDLE EAR & MASTOID · TUMOURS

Adenoma of the Middle Ear

- Evaluation of extension to critical structures such as the internal carotid artery and facial nerve is essential when considering whether complete surgical removal is possible.

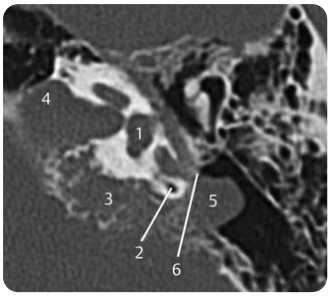


CT, axial. Expansive lesion with lytic destruction of the malleus and incus (1, not clearly shown on this view) as well as anterior bony destruction (2),…

INNER EAR & IAC · TUMOURS

Endolymphatic Sac Tumour

- Signs of bone destruction in the region of the jugular bulb and jugular foramen. –
- Cranial nerve VII XI palsies, extension into structures of the neck and skull base.
- An endolymphatic sac tumor has a high risk of progressive morbidity and surgical complications due to excessive bleeding and cranial nerve damage. Preoperative embolization is recommended if technically possible.
- There is a well-known higher incidence of endolymphatic sac tumor in von Hippel–Lindau disease.
- Endolymphatic sac tumor arising from the middle third of the endolymphatic sac. It is a hypervascular tumor that may show intratumorous calcification. Bone destruction is characteristic, especially in patients with known von Hippel–Lindau disease.
- Remember to consider the possibility of bone destruction in the region of the jugular bulb and jugular foramen and the anticipated clinical symptoms with growth of the tumor when discussing the remaining surgical options with the patient, in particular the effects on cranial nerves VII XI.
- An endolymphatic sac tumor is associated with a high risk of progressive morbidity and surgical complications because of excessive bleeding and cranial nerve damage. Preoperative embolization is recommended if possible.

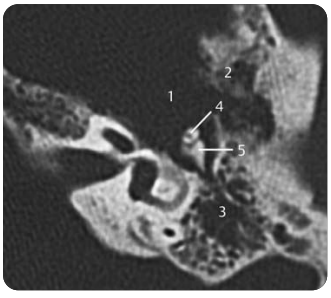


CT, axial. Posterior to the vestibule (1) and posterior semicircular canal (2), an opacification of the mastoid cells with infiltrative characteristics is seen. Thi…

EXTERNAL EAR & TYMPANIC MEMBRANE · TUMOURS

Malignancy of the External Auditory Canal Squamous Cell Carcinoma

- Extension of bony destruction and features suggestive of infiltrative growth are associated with high morbidity, such as dysfunction of the facial nerve and inner ear.
- Adherence to vital structures, such as the internal carotid artery, may preclude surgery.
- Secondary complications, i.e., mastoiditis or otitis media, as a result of stasis of secretions due to obstruction of the eustachian tube. After radiotherapy, necrotizing osteomyelitis and mastoiditis may occur.

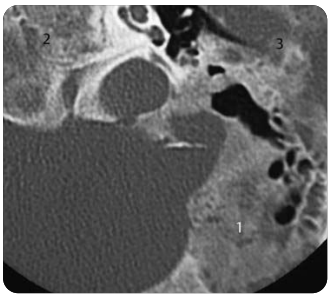


CT, axial. Cranial slice at the level of the horizontal canal, showing destruction of the wall of anterior epitympanum (1), irregular invasion of…

TEMPORAL BONE & SKULL BASE · BONE DISEASE

Fibrous Dysplasia of the Temporal Bone and Skull Base

- Radiologically on CT, enlargement of bone with intact cortical outlines and a dense ground glass appearance is characteristic of fibrous dysplasia. Fibrous dysplasia incidentally found on MRI may be confused with a malignancy.
- In meningioma, bony outlines can be irregular and signs of hyperostosis can be found as well as a dural tail. Sometimes intracranial calcifications might be indicative of an intracranial meningioma.
- In progressive disease, there is encroachment of neural structures and foramina in the skull base, with resultant neurologic and vascular deficits. The external auditory canal may be narrowed and medial to the stenosis; an inclusion cholesteatoma might develop.
- Surgical procedures are limited to prevent complications of encroachment of critical structures and treatment of chronic infections in the middle ear cavity. In obstruction of the auditory canal, extra attention must be paid to the potential risk of inclusion cholesteatoma.
- Fibrous dysplasia in a single osseous structure (monostotic) can be monitored with serial scanning and can be stable and asymptomatic for years.
- In cases of meningioma, the bony outlines can be irregular and hyperostosis can be found as well as a dural tail. Intracranial calcifications may be suggestive of a meningioma.
- In later stages of Paget disease, the otic capsule is completely resorbed. Osteopetrosis demonstrates a complete lack of pneumatization of the temporal bone. Craniometaphyseal dysplasia is recognized by the presence of increased bone densities.
- Beware of encroachment of vital structures and cranial nerves as well as the formation of an inclusion cholesteatoma.



CT, axial. Fibrous dysplasia is characterized by replacement of medullary bone by abnormally proliferating fibrous tissue, resulting i…

TEMPORAL BONE & SKULL BASE · TUMOURS

Pseudotumors in the Apex of the Petrous Bone

IMAGING FINDINGS

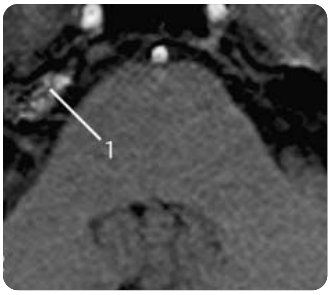
- Asymmetric marrow: bright on T1, suppresses with fat saturation, not expanded
- Trapped fluid: bright on T2, intact septa
- Cholesterol granuloma: bright on T1 and T2, expansile

TOP DIFFERENTIALS

- Cholesteatoma or epidermoid (DWI bright)
- Petrous apicitis (destruction, enhancement)
- Carotid aneurysm

CHECKLIST

- Compare with the other apex
- CT for expansion and cortex



MRI, T1-weighted, axial. T1-weighted image shows that the lesion (1) has high signal intensity compared with brain tissue (2). The contralateral…



Axial bone CT reveals an expansile cholesterol granuloma in the left PA with marginal bone dehiscence present. The right PA is well…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1152.